

Week 2 - Assignment
Data Preprocessing & Exploratory Data Analysis
(EDA)

Iris Dataset

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1. Introduction

The quality of a machine learning model depends on the quality of the data used for training. Before developing predictive models, it is important to preprocess the dataset by handling missing values, checking duplicates, detecting outliers, encoding categorical variables, and scaling numerical features. Exploratory Data Analysis (EDA) is then performed to understand data distributions, identify relationships among variables, and discover useful patterns.

The objective of this assignment is to preprocess the **Iris dataset**, perform exploratory data analysis, and prepare the dataset for future machine learning model development. The target variable for this dataset is **Species**, which represents the type of Iris flower.

2. Dataset Overview

The Iris dataset is one of the most widely used datasets for machine learning classification tasks. It contains measurements of Iris flowers from three different species.

Dataset Information

- **Number of Rows:** 150
- **Number of Columns:** 6
- **Target Variable:** Species

Numerical Features

- Id
- SepalLengthCm
- SepalWidthCm
- PetalLengthCm
- PetalWidthCm

Categorical Feature

- Species

The dataset was inspected using **head()**, **info()**, **describe()**, **isnull()**, and **duplicated()** to understand its structure and identify any data quality issues before preprocessing.

3. Initial Data Inspection

The initial inspection showed that the dataset contains **150 rows** and **6 columns**. There were **no missing values** and **no duplicate records**. The dataset consists of one identifier column, four numerical features, and one categorical target variable.

Overall, the dataset was well-structured and required only minimal preprocessing before machine learning preparation.

4. Data Preprocessing

- Missing Values

```
df.isnull().sum()

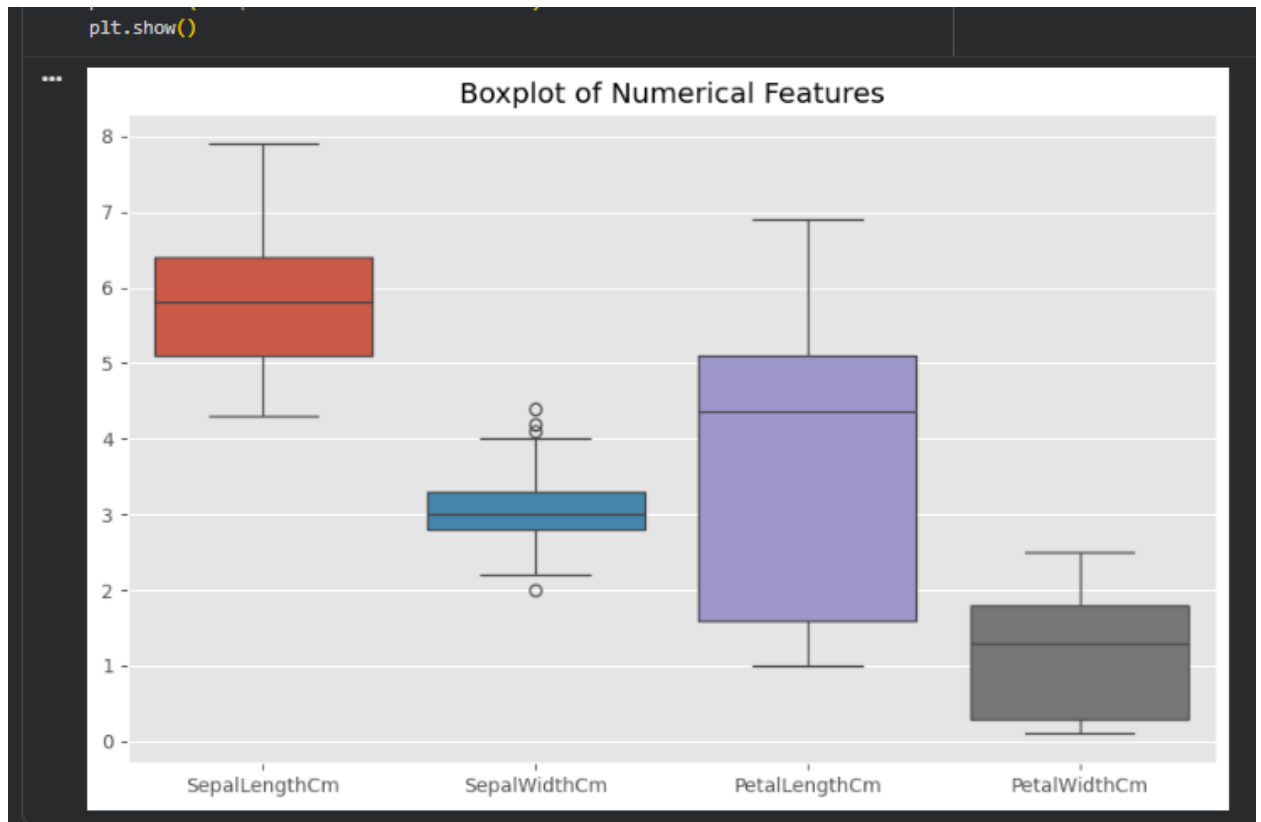
0
Id      0
SepalLengthCm  0
SepalWidthCm  0
PetalLengthCm  0
PetalWidthCm  0
Species    0

dtype: int64
```

- Duplicate Check

```
df.duplicated().sum()
np.int64(0)
```

- Boxplots



-
- Scaled Dataset

Next steps: [Generate code with df](#) [New interactive sheet](#)

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	-0.900681	1.032057	-1.341272	-1.312977	0
1	-1.143017	-0.124958	-1.341272	-1.312977	0
2	-1.385353	0.337848	-1.398138	-1.312977	0
3	-1.506521	0.106445	-1.284407	-1.312977	0
4	-1.021849	1.263460	-1.341272	-1.312977	0

Data Preprocessing Summary

Task	Method Used	Reason
Missing Values	None Required	No missing values were found in the dataset.
Duplicate Records	None Found	No duplicate records existed.
Outliers	Retained	Outliers represented natural biological variation.
Id Column	Removed	It is an identifier and has no predictive value.
Species	Label Encoding	Converted categorical classes into numerical values.
Feature Scaling	StandardScaler	Standardized numerical features for machine learning.

Assignment Questions

Q: Why was no missing value treatment required?

Answer:

The Iris dataset contained **no missing values**, so no imputation techniques such as mean, median, or mode replacement were necessary.

Q: Why were duplicate records not removed?

Answer:

The dataset contained **no duplicate records**, so no data cleaning was required for duplicate observations.

Q: Why were outliers retained?

Answer:

A few outliers were observed in **SepalWidthCm**, but they represent genuine biological measurements rather than data entry errors. Therefore, they were retained to preserve the natural variation within the dataset.

Q: Why was the Id column removed?

Answer:

The **Id** column is only a unique identifier and does not contribute to predicting the flower species. It was removed to prevent irrelevant information from influencing the machine learning model.

Q: Why was Label Encoding used?

Answer:

The **Species** column contains three categorical classes. Label Encoding converted these classes into numerical values (0, 1, and 2), making the target variable suitable for machine learning algorithms.

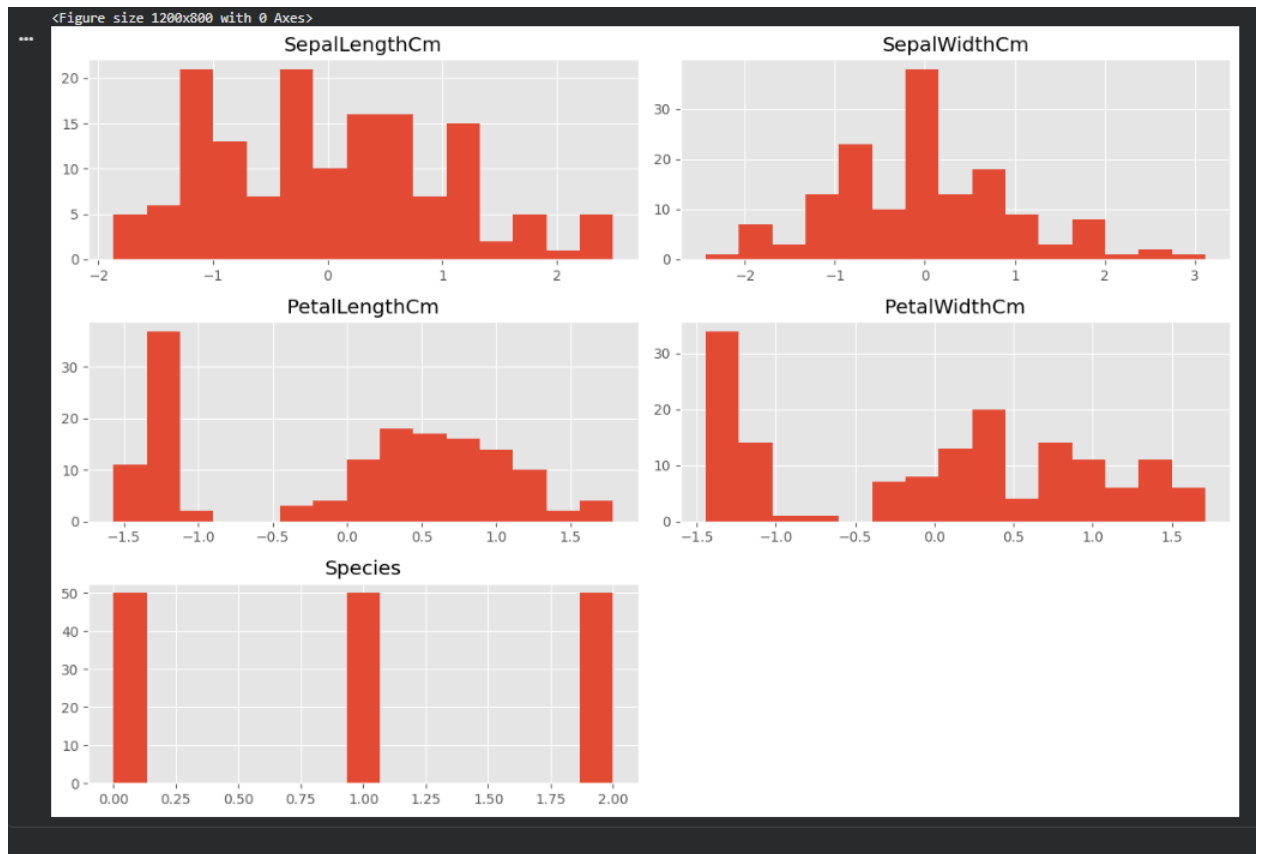
Q: Why was StandardScaler applied?

Answer:

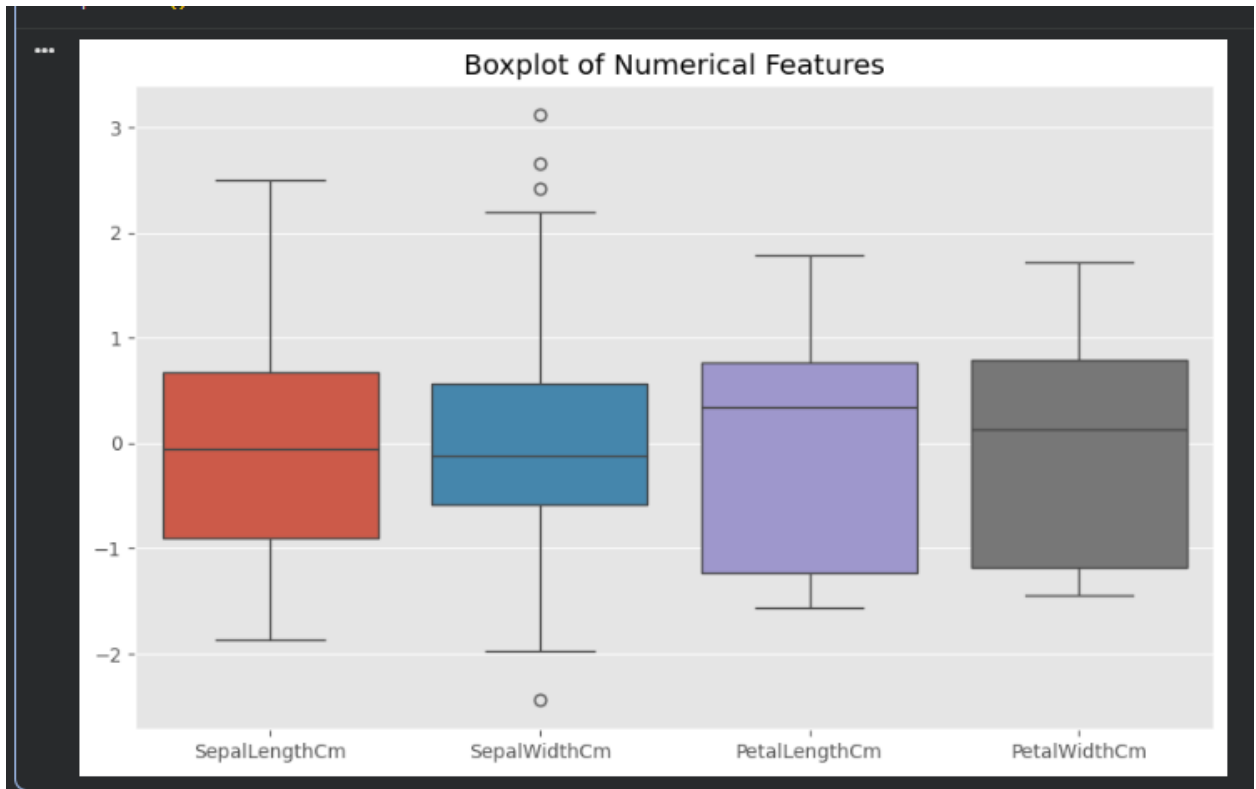
StandardScaler standardizes numerical features by transforming them to have a mean of 0 and a standard deviation of 1. This ensures that all features contribute equally during machine learning model training.

5. Exploratory Data Analysis (EDA)

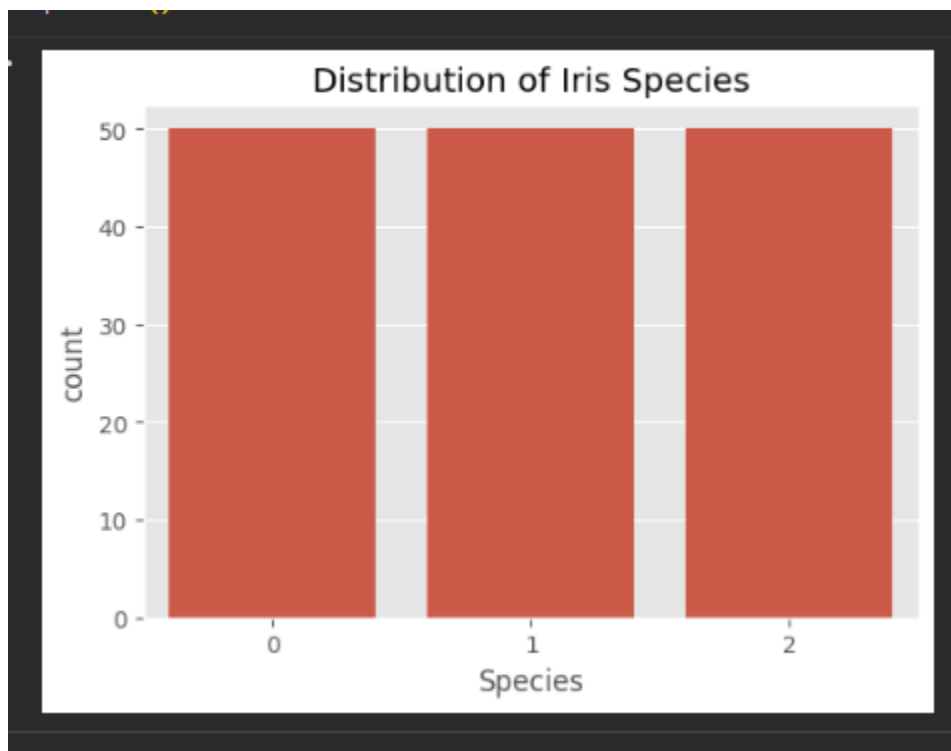
- **Histogram**



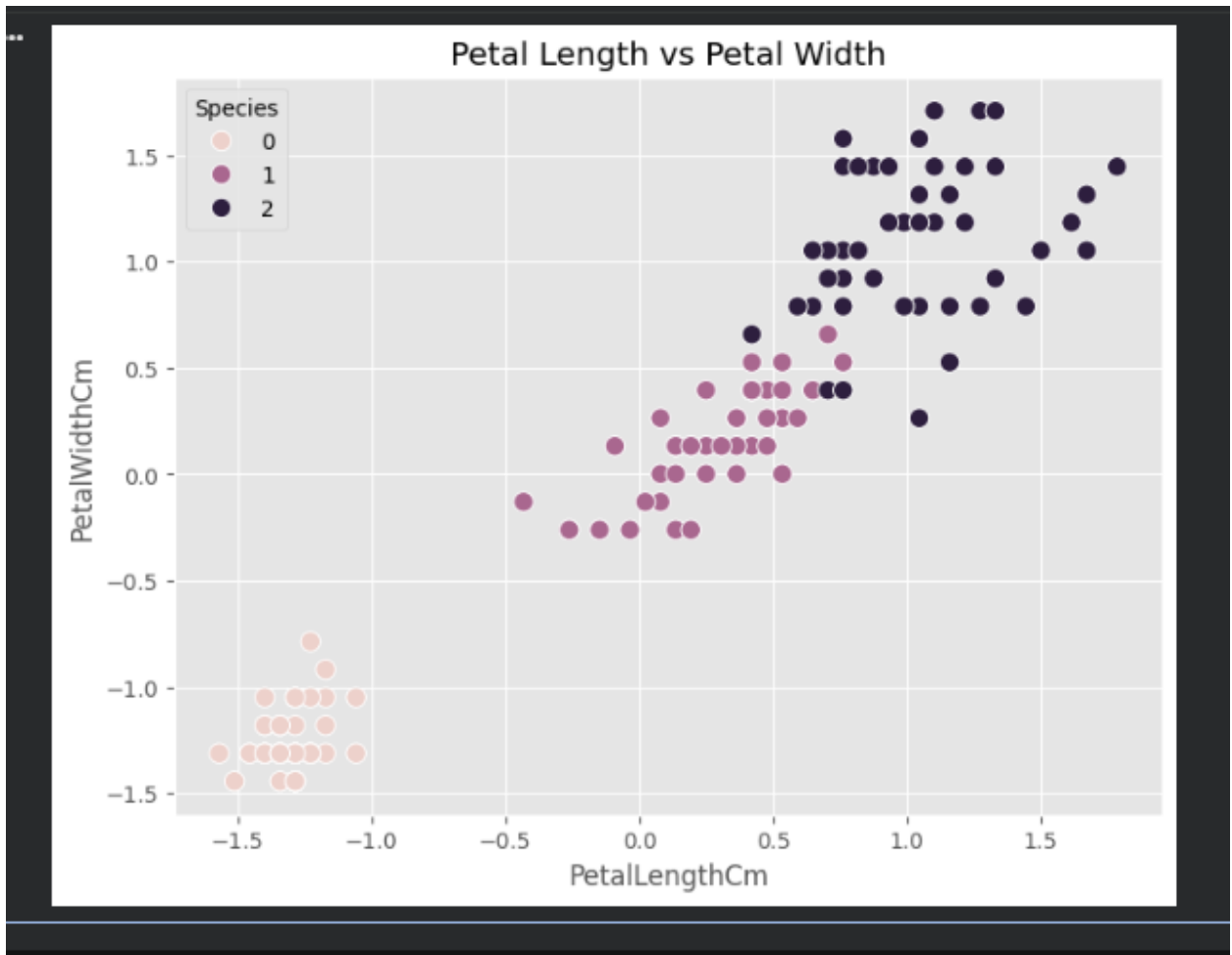
- **Boxplot**



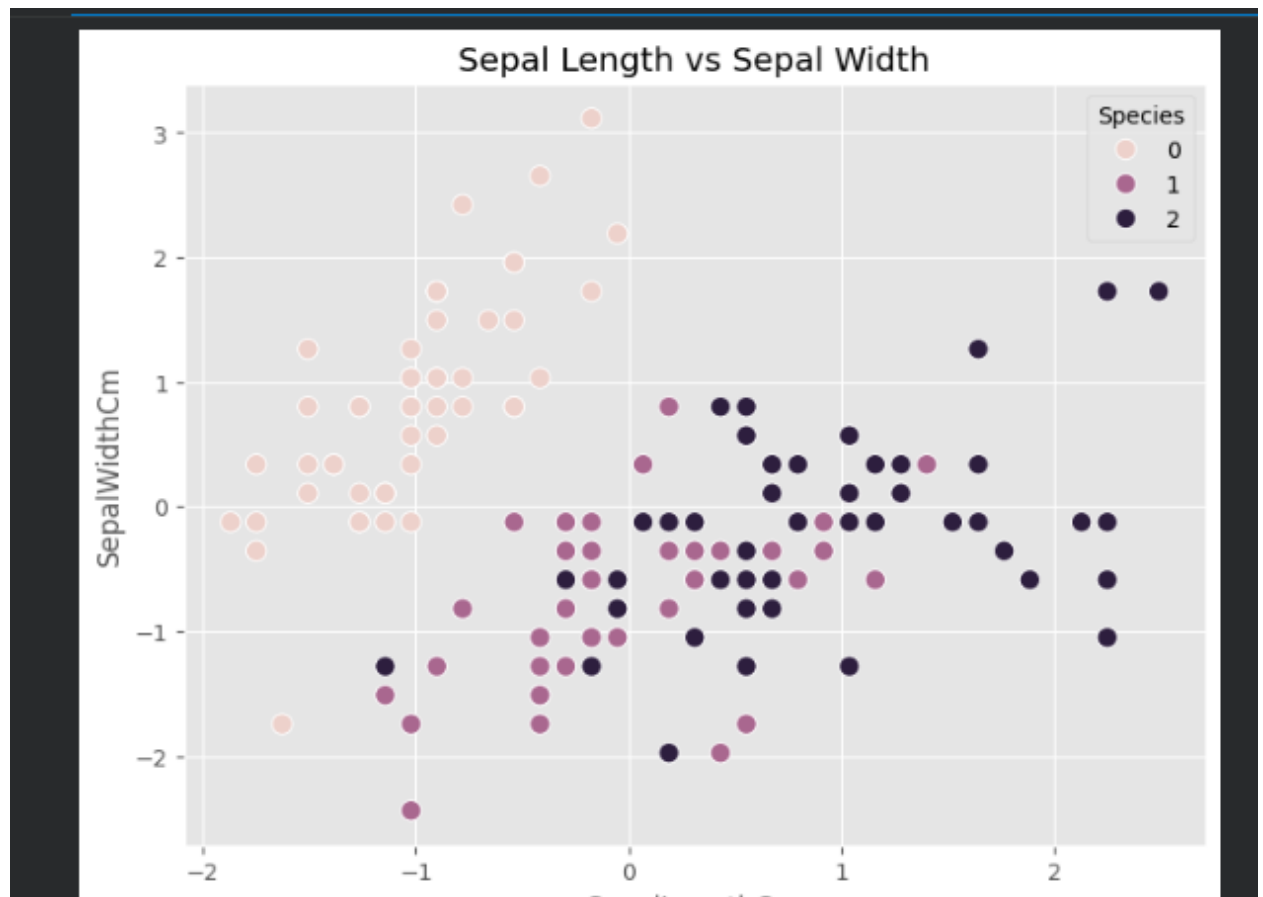
-
- **Species Count Plot**



- **Petal Length vs Petal Width (Scatter Plot)**



-
- **Sepal Length vs Sepal Width (Scatter Plot)**



Histogram Analysis

The histograms were used to examine the distribution of the numerical features. **SepalWidthCm** approximately follows a normal (bell-shaped) distribution, while **PetalLengthCm** and **PetalWidthCm** show a bimodal distribution. This indicates that petal measurements vary significantly across the three Iris species and are useful for classification.

Boxplot Analysis

The boxplots summarize the spread of the numerical features and highlight potential outliers. A few outliers were observed in **SepalWidthCm**, while the remaining features showed little or no significant outliers. These values were retained because they represent natural biological variation.

Species Count Plot

The count plot shows that each Iris species contains **50 observations**, resulting in a perfectly balanced dataset. A balanced dataset helps machine learning models learn all classes equally and reduces prediction bias.

Petal Length vs Petal Width

The scatter plot shows a **strong positive relationship** between petal length and petal width. The three Iris species form distinct clusters, making these features highly effective for distinguishing flower species.

Sepal Length vs Sepal Width

The scatter plot indicates a weaker relationship between sepal length and sepal width. While **Species 0** forms a relatively distinct cluster, **Species 1** and **Species 2** overlap considerably, making sepal measurements less effective for classification.

Assignment Questions

Q: Which feature has the highest variation?

Answer:

PetalLengthCm has the highest variation because its values are spread across a wider range than the other numerical features.

Q: Are there any skewed distributions?

Answer:

SepalWidthCm is approximately normally distributed, while **PetalLengthCm** and **PetalWidthCm** show a bimodal distribution due to differences among the three Iris species.

Q: Which category occurs most frequently?

Answer:

All three Iris species occur equally, with **50 observations** each, indicating a balanced dataset.

Q: Which variables are most related to the target variable?

Answer:

PetalLengthCm and **PetalWidthCm** show the strongest relationship with the target variable (**Species**) and provide the clearest separation between the flower classes.

Q: What trends were observed?

Answer:

- Petal length and petal width increase together, showing a strong positive relationship.
- The three Iris species form distinct clusters based on petal measurements.
- Sepal measurements show greater overlap, particularly between Species 1 and Species 2.
- Species 0 is clearly distinguishable due to its smaller petal size.
- Petal features are more effective for species classification than sepal features.

6. Correlation Analysis

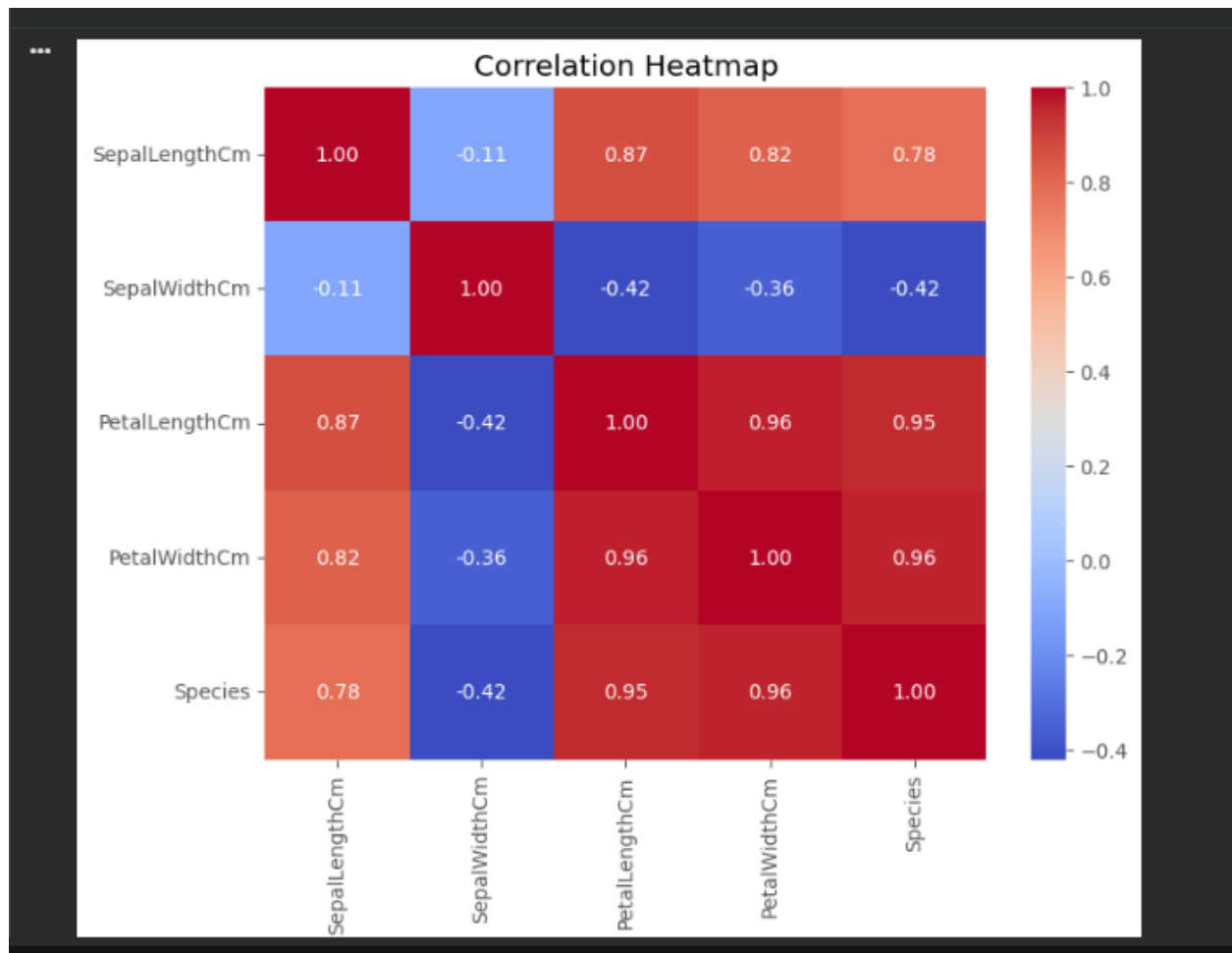
Start coding or generate with AI.

Index	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
SepalLengthCm	1.0	-0.1093692499506494	0.8717541573048717	0.8179536333891635	0.7825612310100813
SepalWidthCm	-0.1093692499506494	1.0	-0.4205160964011546	-0.35654400961380557	-0.419446200260027
PetalLengthCm	0.8717541573048717	-0.4205160964011546	1.0	0.9627570970509685	0.9490425448522336
PetalWidthCm	0.8179536333891635	-0.35654400961380557	0.9627570970509685	1.0	0.9564938238016174
Species	0.7825612310100813	-0.419446200260027	0.9490425448522336	0.9564938238016174	1.0

Show 25 rows per page.
Like what you see? Visit the [data lake notebook](#) to learn more about interactive tables.

Next steps: [Generate code with corr](#) [New interactive sheet](#)

```
plt.figure(figsize=(8,6))  
sns.heatmap(corr,
```



The correlation matrix and heatmap were generated to examine the relationships among the numerical features in the Iris dataset. The analysis showed that **PetalLengthCm** and **PetalWidthCm** have the strongest positive correlation, indicating that flowers with longer petals generally have wider petals. **SepalLengthCm** also has a positive relationship with petal measurements, while **SepalWidthCm** shows weaker correlations with the other features.

No strong negative correlations were observed. The results indicate that petal measurements are the most informative features for classifying Iris flower species.

7. Machine Learning Readiness Summary

The Iris dataset has been successfully prepared for machine learning model development.

The following preprocessing steps were completed:

1. No missing values were found in the dataset.
2. No duplicate records were identified.
3. The **Id** column was removed because it had no predictive value.
4. Outliers in **SepalWidthCm** were retained as they represent natural biological variation.
5. **Label Encoding** was applied to convert the **Species** column into numerical values.
6. **StandardScaler** was used to standardize all numerical features.

After preprocessing, the dataset became clean, encoded, standardized, and ready for machine learning model development.

8. Key Insights

1. The Iris dataset contains **150 flower samples** equally distributed across three species.
2. **PetalLengthCm** and **PetalWidthCm** are the strongest features for distinguishing flower species.
3. The dataset contains **no missing values** or **duplicate records**, indicating high data quality.

4. A few natural outliers were found in **SepalWidthCm**, but they were retained because they represent genuine observations.
5. After preprocessing, the dataset is fully prepared for machine learning classification tasks.

9. Conclusion

This project successfully demonstrated the complete data preprocessing and exploratory data analysis workflow for the Iris dataset. The dataset was inspected, cleaned, encoded, and standardized using appropriate preprocessing techniques.

Exploratory data analysis revealed that petal-related features are the most important variables for distinguishing between Iris species, while sepal measurements provide comparatively less separation. The final cleaned dataset is free from data quality issues and is fully prepared for machine learning model development in future stages of the internship.